In [1]:

```
import pandas as pd

data={
    'Income':[15,16,17,50,52,80,82,85],
    'Spending':[20,22,18,55,58,80,85,82]
}
```

In [5]:

```
df=pd.DataFrame(data)
```

In [7]:

```
df
```

Out[7]:

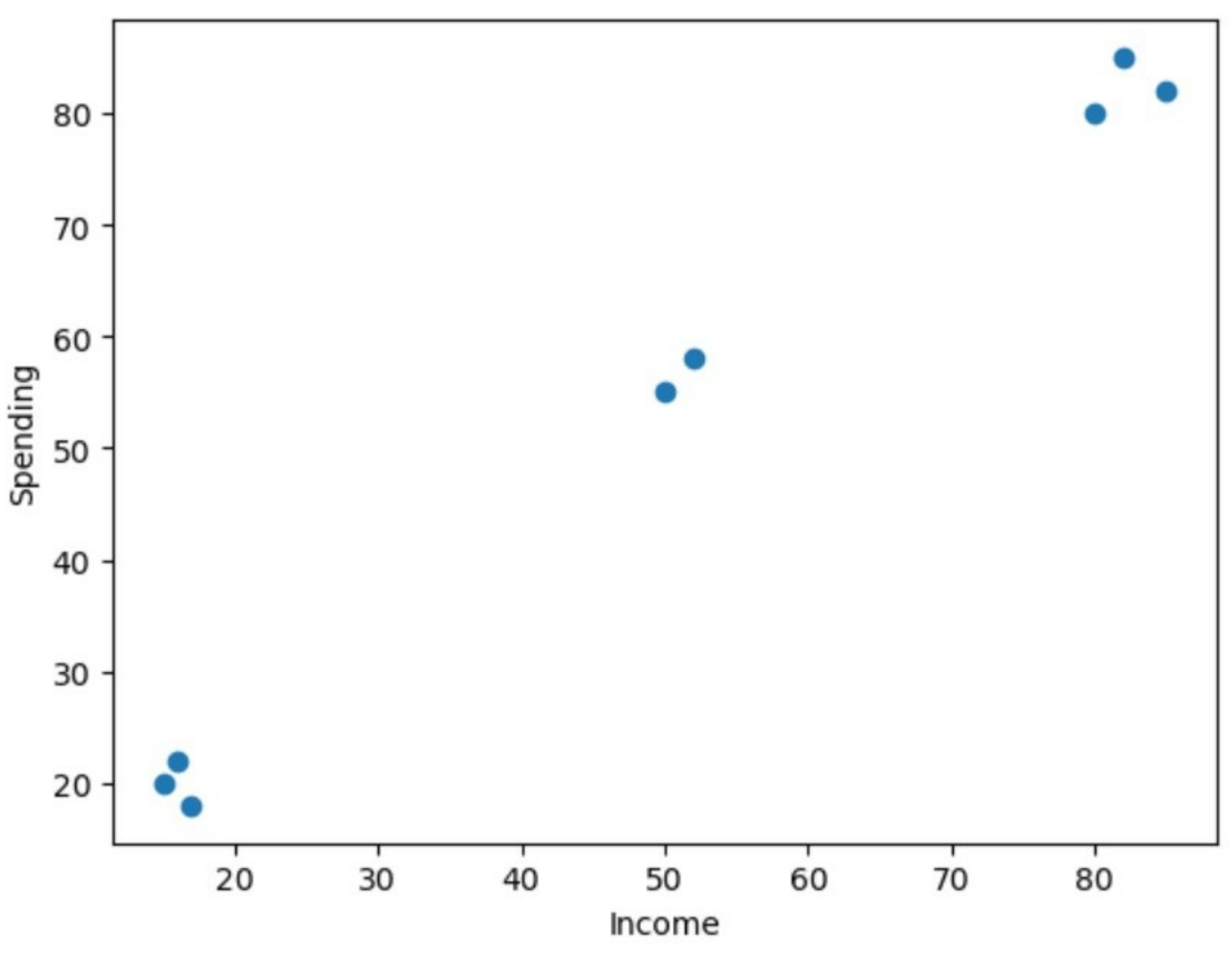
	Income	Spending
0	15	20
1	16	22
2	17	18
3	50	55
4	52	58
5	80	80
6	82	85
7	85	82

In [9]:

```
import matplotlib.pyplot as plt

plt.scatter(
    df['Income'],
    df['Spending']
)

plt.xlabel('Income')
plt.ylabel('Spending')
plt.show()
```



Feature scaling

In [12]:

```
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()

x=sc.fit_transform(df)
```

In [14]:

```
x
```

Out[14]:

```
array([[-1.20349499, -1.19715289],
       [-1.16873701, -1.12348194],
       [-1.13397903, -1.27082383],
       [ 0.01303424,  0.09208868],
       [ 0.0825502 ,  0.2025951 ],
       [ 1.05577358,  1.01297552],
       [ 1.12528954,  1.19715289],
       [ 1.22956347,  1.08664647]])
```

Create Dendogram

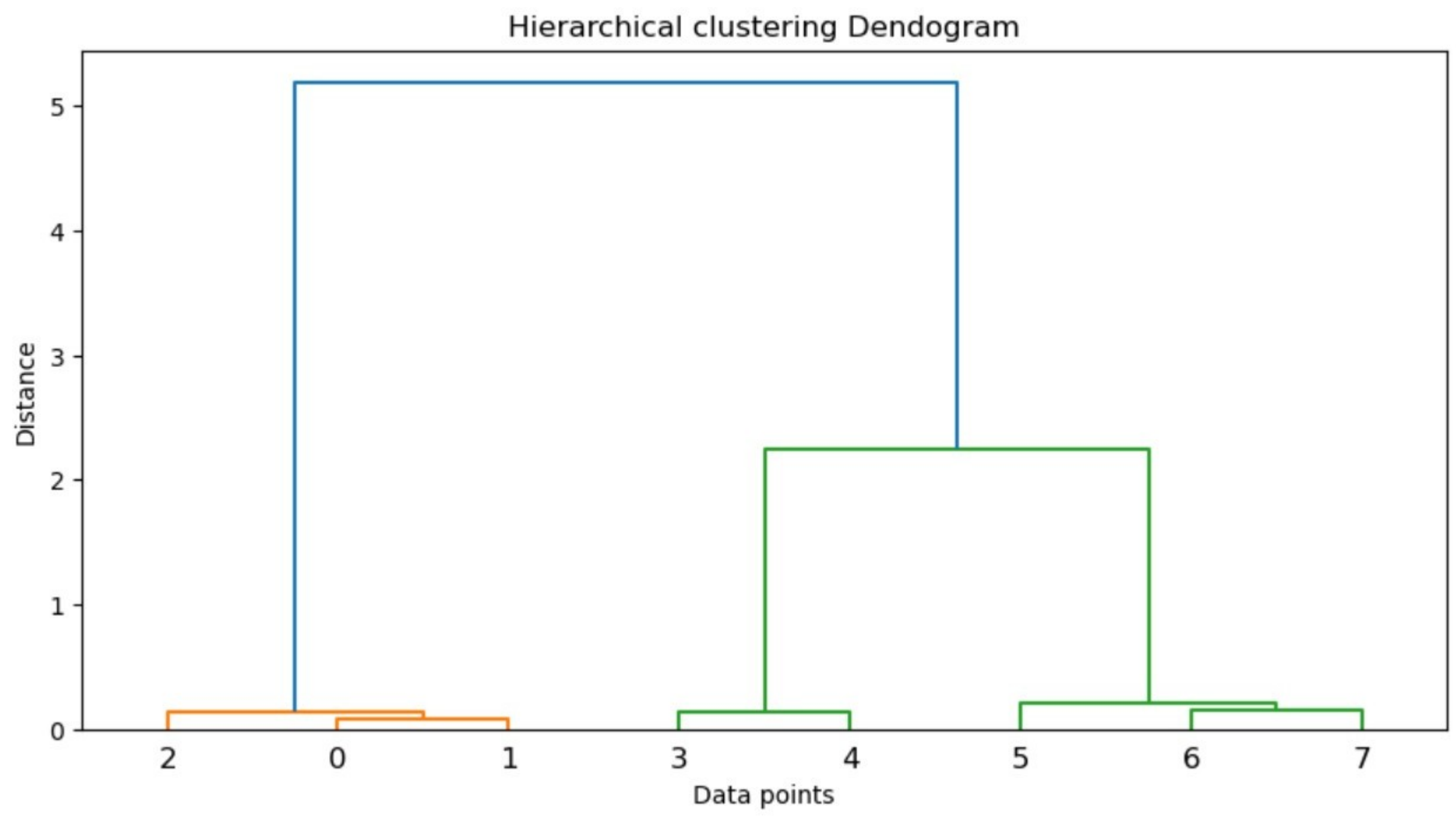
In [25]:

```
from scipy.cluster.hierarchy import dendrogram,linkage

z=linkage(x,method='ward')

plt.figure(figsize=(10,5))

dendrogram(z)
plt.xlabel("Data points")
plt.ylabel("Distance")
plt.title("Hierarchical clustering Dendogram")
plt.show()
```



Model

In [55]:

```
from sklearn.cluster import AgglomerativeClustering
model=AgglomerativeClustering(
    n_clusters=3,
    linkage='ward'
)

labels=model.fit_predict(x)
print(labels)
```

```
[1 1 1 2 2 0 0 0]
```

In [57]:

```
df['clusters']=labels
```

In [59]:

```
df
```

Out[59]:

	Income	Spending	clusters
0	15	20	1
1	16	22	1
2	17	18	1
3	50	55	2
4	52	58	2
5	80	80	0
6	82	85	0
7	85	82	0

In [61]:

```
plt.figure(figsize=(5,5))

plt.scatter(
    df['Income'],
    df['Spending'],
    c=df['clusters']
)
plt.xlabel('Income')
plt.ylabel('spending')
plt.title('Customer segmentation')
```

Out[61]:

```
Text(0.5, 1.0, 'Customer segmentation')
```